

Random Walks

Text

For example, the visualization here shows 3 node network, and the associated transition matrix. The worker is initially at node A. Note that the transition probability from node C, to node A is 50%. While the transition probability from node A to node C is 0. An important question is, what is the probability that the work here will be found at the given node after many such transitions? If these probabilities converge asymptotically, we can think of them as the fraction of time that the worker spends at its node. This is referred to as the stationary distribution, of the random work. An important result is that in undirected and connected networks, the stationary distribution always exists. The visualization, shows the stationary distribution for these random work. And as you can see, the fraction of time that the worker is expected to be at node B is equal to the fraction of time it is expected to be at node C. And the only half of that time of node A.
